

SmritiSetu — Cognitive Gaming Module

Full Design Document: Games, Research Grounding & Voice Integration

This document covers ONE part of the larger project — the gaming and cognitive training module. It stands as its own mini-project because it is the single largest piece of engineering and research work in the whole platform. Every game listed here is backed by either a real published study, a documented clinical task, or a validated commercial serious-game precedent — nothing here is invented from imagination.

Part 1 — Why We Designed It This Way (The Thinking Behind the Whole Module)

Before listing the games, it's worth writing down *how* we arrived at this list, because a judge or reviewer will ask "why these games and not others?"

The core decision: don't build one game reskinned four times. The problem statement asks for four ability categories — memory, attention, daily routine recall, and pattern recognition. It would be easy (and lazy) to build one matching-pairs game and just swap the pictures four times to "cover" all four categories. We deliberately avoided this. Real cognitive science treats memory, attention, and executive function as separate mental abilities that need separate types of tasks to train — so we picked a different task *mechanic* for each ability, not just a different skin.

The second decision: don't invent tasks — adapt tasks that already exist and have been tested. Every game below is built on top of either:

A task type used in an actual published research study or clinical trial, or

A task type used in a real, currently-selling serious game for dementia/cognitive decline

We are not claiming our version has been clinically proven — that would be dishonest, since we haven't run a trial. What we ARE claiming is: *this exact type of task has been tested before, on similar patients, and has shown real, published results — we are rebuilding that same task, in Assamese, with culturally familiar content, for our specific users.* That is an honest, defensible claim.

The third decision: no timers, no red "wrong" marks, no failure state. Multiple sources we studied (Memory Lane Games' design philosophy, published serious-game design guidance) specifically warn that speed pressure and harsh failure feedback increase anxiety in dementia patients and reduce engagement rather than helping. So difficulty in our games is scaled by *how much information the person has to hold or process*, not by *how fast they must respond*. Every wrong answer gets a gentle, encouraging response, never a buzzer or red cross.

The fourth decision: real content, not stock content. Instead of generic shapes or Western-style photos, every game uses real Assamese cultural material — Bihu, local food, local objects, everyday routines familiar to someone who has lived in Assam their whole life. This isn't just a nice-to-have; recognition-based memory (does this feel familiar to *me*) works better than abstract recall for people with cognitive decline, which is exactly why reminiscence-therapy apps use personal/cultural photos as their core mechanic.

Part 2 — The Full Game Arsenal (10 Games, One by One)

For every game below, we explain: what ability it targets, why we say it targets that ability, what real-world research or product it is based on, and exactly what we would build.

GAME 1: "Ki Utsav?" (What Festival?) — Episodic Memory / Reminiscence

What ability it targets: Episodic memory — the ability to recall specific past events, experiences, and personally/culturally meaningful moments.

Why we say it targets this ability: Episodic memory is one of the first types of memory to weaken in early cognitive decline, but long-held cultural and life memories (formed decades ago) tend to stay intact much longer than recent memories. Reminiscence-based recall tasks are specifically designed to reach into that older, more stable memory store rather than testing short-term memory, which is unfair to test in this population.

Research basis: A serious game built specifically to target episodic memory used an "iPad Game Show" format aimed directly at addressing motivational deficits in older adults with memory impairment. Separately, commercial reminiscence-therapy apps (Memory Lane Games, AmuseIT) are built entirely around this same mechanic — showing an image and asking a recall or recognition question, explicitly avoiding "wrong answer" framing.

What we build: A screen shows a photo — a Bihu celebration, a harvest scene, a common festival moment. The avatar asks, out loud in Assamese, a simple question like "What festival is this?" or "What do people usually eat during this time?" The person answers by tapping one of three large picture options, or by speaking the answer aloud. There is no penalty for a wrong answer — the avatar gently confirms the correct one and moves on. Over time, the app tracks (silently, never shown to the patient) how often recall is correct and how much prompting was needed.

GAME 2: "Ki Ki Silo?" (What Were They?) — Working Memory

What ability it targets: Working memory — the ability to briefly hold a small number of items in mind and recall them correctly a short time later.

Why we say it targets this ability: This is a direct, simplified version of a task type used to measure short-term/working memory in real cognitive assessments — hold a small set of items, then recall them after a brief distraction. This is different from episodic memory because the "event" being remembered was invented seconds ago by the game, not decades ago by real life.

Research basis: Working memory is explicitly listed as one of the trainable cognitive domains in CogniFit's validated 21-task computerized cognitive training program, which covers 14 distinct cognitive domains including working memory and visual short-term memory.

What we build: The screen briefly shows 3 to 4 everyday objects (a cup, a gamosa, a mustard-oil bottle, a betel leaf) for a few seconds. They disappear. The person is then asked, "Which of these did you see?" from a slightly larger set of pictures, including a couple that weren't shown. Difficulty increases not by adding a timer, but by increasing the number of items shown or making the "distractor" objects more visually similar to the real ones.

GAME 3: "Kot Gol?" (Where Did It Go?) — Spatial / Direction Memory

What ability it targets: Spatial memory — remembering a location, path, or sequence of movement.

Why we say it targets this ability: Spatial memory is a distinct cognitive skill from object memory or event memory — it's about "where," not "what" or "when." Difficulty with spatial memory (getting lost, forgetting the layout of a familiar place) is a commonly reported early symptom in cognitive decline, making it a meaningful skill to specifically track and gently train.

Research basis: "Direction Memorizing" is one of five validated game types included in a published mobile brain-training platform designed for older adults with mild cognitive impairment, specifically built to target spatial processing as one of several distinct cognitive domains.

What we build: A simple animated story — for example, a small boat moving down a river, or a person walking a path through a village — plays out on screen with narration. Afterward, the person is asked to recall or point out the path taken ("did the boat turn left or right at the tree?"). Kept extremely simple and short for MVP — 2-3 steps in the path, not a maze.

GAME 4: "Etu Cabo!" (Look at This!) — Sustained Attention

What ability it targets: Sustained attention — the ability to stay focused on a task over a period of time without drifting.

Why we say it targets this ability: This is a direct simplification of a well-established, validated clinical task: the "detection task," where a person is asked to respond as soon as a specific visual stimulus appears. This exact task format is used in real computerized cognitive test batteries to measure basic attention and information-processing speed.

Research basis: The detection task (press a button as soon as a card turns face-up) is used in a well-validated simple reaction-time paradigm within a real clinical cognitive assessment battery. A separate 2026 randomized controlled trial using a mobile attention-training game called "Descartes" found a statistically significant improvement in the attention domain of the MoCA (a standard cognitive assessment) compared to a control group.

What we build: A calm scene (a pond, a market stall) is shown, and every so often, a specific object appears (a duck surfacing, a customer arriving). The person taps the screen whenever they see it appear. No timer countdown is shown, and there's no penalty for missing one — it's a gentle, ongoing awareness exercise, not a speed test.

GAME 5: "Bhinno Ke?" (Which Is Different?) — Selective Attention

What ability it targets: Selective attention — the ability to focus on the relevant detail while ignoring similar, distracting information.

Why we say it targets this ability: This is different from sustained attention (Game 4) because the challenge here isn't staying alert over time — it's picking out one correct detail from several similar-looking options, which requires actively filtering out distractors.

Research basis: This mirrors the "identification task" used in a real validated clinical cognitive battery, where a person must decide whether a specific card matches a target condition (e.g., "is it red?") among visually similar options — a well-established measure of visual/selective attention.

What we build: A row of similar objects is shown (several gamosas, mostly one color, one a different color) and the person is asked to tap the one that's different, or the one matching a description spoken by the avatar. Difficulty scales by making the "different" item less obviously different over time — not by adding time pressure.

GAME 6: "Loga Loi Jua" (Take Them Along) — Daily Routine Recall / Sequencing

What ability it targets: Procedural memory and executive function — the ability to remember and correctly order the steps of a familiar daily task.

Why we say it targets this ability: This is arguably the most important game category for real caregivers, because it mirrors actual Activities of Daily Living (ADL) — the real-world tasks whose breakdown is one of the clearest signs of worsening cognitive decline, and one of the most disruptive to independence.

Research basis: A published point-and-click serious game used real cooking scenarios (toasting bread, making coffee) specifically to train planning ability and object recognition together, grounded in the idea that everyday, ecologically valid tasks transfer better to real life than abstract puzzle-style games.

What we build: The person is shown 4-5 picture cards representing steps of a familiar routine — making saah (tea), preparing for Bihu, getting ready in the morning — shown out of order. They drag or tap the cards into

the correct sequence, guided by voice narration if needed. Wrong order simply lets them try again with a gentle nudge, no penalty.

GAME 7: "Bosarot Ki Lage?" (What Do We Need?) — Executive Function / Planning

What ability it targets: Executive function, specifically planning and list-based reasoning — deciding what's needed for a task and tracking multiple items at once.

Why we say it targets this ability: Planning is a distinct executive skill from simple recall — it requires holding a goal in mind (what am I preparing for?) and matching several items against that goal, which is a heavier cognitive load than a single memory-recall task.

Research basis: A published serious game targeting exactly this ability used a shopping-list mechanic, where the person selects the correct items needed for a task from a larger set, and explicitly scaled difficulty by adjusting the number of items on the list and the amount of time allowed — chosen by the researchers specifically because it reflects a real activity of daily living, not an abstract test.

What we build: The avatar says, "We're preparing for guests coming for tea — what do we need?" and shows a large set of pictured items (tea leaves, cups, biscuits, sugar, but also irrelevant items like a broom or a lamp). The person selects only the relevant ones. Difficulty scales by increasing the total number of items shown, exactly as in the cited research.

GAME 8: "Milai Diya" (Match Them) — Pattern & Object Recognition

What ability it targets: Visual pattern recognition and object-matching ability.

Why we say it targets this ability: Matching requires comparing visual features (shape, color, category) across multiple items and identifying which belong together — a foundational visuospatial skill that's simpler and more forgiving than recall-based tasks, making it a good "easy win" game for confidence-building.

Research basis: Picture-matching and picture-observation games (identifying colors, positions, or content of images) are among the earliest and most consistently used task types in dementia-focused serious games, cited specifically as tools for cognitive screening and gentle training in early-stage cognitive change.

What we build: Classic pair-matching, but using real Assamese cultural objects — a jaapi, a xorai, a gamosa, local fruits — instead of generic shapes or cartoon icons. Grid size (4 pairs vs 6 pairs) is the difficulty lever, not speed.

GAME 9: "Bak Kotota?" (How Many?) — Attention + Processing Speed

What ability it targets: Sustained attention combined with basic numeric processing — counting under mild distraction.

Why we say it targets this ability: Counting tasks are a simple, low-stakes way to measure whether a person can maintain focus while processing a small amount of numeric information — useful because numeric processing difficulty is often an early, subtle sign of cognitive change that's easy to gently monitor through a game.

Research basis: "Rabbit Counting" and "Triangle Counting" are validated components of a published, tested mobile brain-training platform explicitly designed to target attention and processing speed together, alongside working memory and spatial tasks.

What we build: A simple animated scene (ducks crossing a pond, boats on a river) where a small number of objects appear, and the person is asked "how many ducks did you see?" Kept to very small numbers (2 to 5) for MVP, since the goal is engagement and gentle monitoring, not a hard numeracy test.

GAME 10: "Ki Koribo?" (What Would You Do?) — Simple Decision-Making

What ability it targets: Basic decision-making and executive response — choosing between two or three simple options in a low-stakes scenario.

Why we say it targets this ability: Decision-making tasks add a layer beyond memory and attention — they measure whether the person can weigh a small choice and commit to a response, which is a gentler, everyday-relevant version of executive function testing.

Research basis: A Rock-Paper-Scissors-style decision task is included as a validated component of the same published five-game mobile brain-training platform used for several of the games above, chosen specifically because it's simple, familiar, and has an immediate, low-stakes outcome.

What we build: Simple two-option or three-option "what would you do" scenarios grounded in daily life ("it's raining, what would you carry — an umbrella or a fan?"), spoken and shown visually, with warm, encouraging feedback regardless of the answer chosen.

Part 3 — Honest Note on What the Evidence Actually Shows

We want to be upfront rather than oversell this. The published research on cognitive games for dementia is genuinely mixed:

Some studies show **real, measurable benefits** — one study found 16 weeks of combined cognitive-physical game training significantly improved working memory and executive function; another found computerized cognitive training was associated with increased gray matter volume in the brain.

Other reviews found generic "brain games" showed **no significant improvement** in memory or cognitive function at all.

The pattern that emerges across the literature is that **purpose-built, ecologically valid games (mirroring real daily tasks) tend to outperform generic, abstract brain-training games** — which is exactly why every game above is built around a real daily-life task or a validated clinical task type, not an arbitrary puzzle.

For people already diagnosed with dementia (not just early mild cognitive impairment), the research is even more specific: the clearest, most meaningful benefits were found in interventions using **immersive or movement-based technology** (VR, motion-based games), not flat screen-tap games alone. This is an important scoping note — our MVP is screen-based, so we should be honest that this fits best for early-stage / MCI users, with movement-based expansion as a possible future direction, not a claim we make now.

What we do NOT claim: that this app cures or reverses dementia, or guarantees memory improvement.

What we DO claim: these are the same task types used in real, published cognitive-training research, adapted respectfully and specifically for Assamese elderly users — offering meaningful cognitive engagement, monitoring, and stimulation, consistent with what the evidence supports.

Part 4 — Voice Models: What We're Using and Why

Which models, and why these specifically

We use a **hybrid combination of two Government of India-backed, open, Assamese-supporting voice AI systems: Bhashini** (the national language AI mission) and **AI4Bharat's IndicTTS/IndicConformer models** (the research lab whose models power much of Bhashini itself).

Why these and not a foreign commercial provider (Google, Amazon, etc.):

Assamese is explicitly and natively supported by both — confirmed directly by checking Bhashini's own published service list, which names Assamese for ASR, TTS, and translation.

They are free for non-commercial use and government-backed, which matters for both cost and for alignment with an MDoNER (Ministry of Development of North Eastern Region) submission — using another Government of India digital initiative strengthens the feasibility and policy-alignment story of our pitch. Commercial voice AI providers have very limited or no support for Assamese specifically — most Indic voice AI products focus on Hindi and a handful of major south Indian languages first, leaving low-resource languages like Assamese underserved. Bhashini/AI4Bharat is currently the most realistic, production-usable option for this specific language.

What each part of the voice system actually does

Text-to-Speech (TTS) — converts written text into spoken audio. This is the *stronger, more reliable* half of the system. It is used for the avatar's spoken instructions, encouragement, and reminders. Under the hood, this uses a two-stage pipeline: an "acoustic model" (AI4Bharat's FastPitch, or in newer releases, an end-to-end VITS-based model) turns text into a kind of sound blueprint, and a "vocoder" (HiFi-GAN) turns that blueprint into actual playable audio.

Automatic Speech Recognition (ASR) — converts the person's spoken voice into text the app can understand. This is the *weaker, riskier* half — published research shows Assamese ASR error rates are meaningfully higher than Hindi or English (in some studies, around 19-30% character error rate), because far less training data exists for Assamese compared to bigger languages. We are honest about this limitation rather than pretending it isn't there.

How this actually helps and personalizes each game

Because ASR isn't fully reliable for open, free-form speech in Assamese yet, **we don't rely on it to understand anything the person might say — only to recognize whether their answer matches one of a small number of expected options for that specific game screen.** This is a much easier and more reliable task than general speech understanding, and it's what makes voice interaction usable today rather than something we have to wait years for.

Concretely, per game:

In "Ki Utsav?" (Game 1), when the avatar asks "what festival is this," it's only checking whether the spoken answer matches one of the 3 shown picture options — not understanding open conversation.

In "Loga Loi Jua" (Game 6), voice is used mainly one-directionally — the avatar narrates the steps aloud so a person with reading difficulty can still follow the sequencing task, removing the need to read on-screen text at all.

Across every game, TTS is what makes the game accessible to someone who may not read comfortably — every instruction, every question, every piece of encouragement is spoken aloud, not just displayed as text.

Personalization angle: Because the caregiver dashboard allows a family member to type in personalized content (a specific memory, a specific relative's name, a custom question), that personalized text needs *live* TTS synthesis (since it can't be pre-recorded — it didn't exist until the caregiver typed it), while all the fixed, repeating instructions and feedback phrases across every game are recorded once and reused, keeping the app fast, offline-capable, and low-cost for the 90% of interactions that are predictable.

Part 5 — Innovation Layer: What We're Adding Beyond the Existing Research

To be clear about what is "proven precedent" versus what is "our own addition":

Language and cultural localization itself is the core innovation. None of the cited research or products (CogniFit, Memory Lane, the five-game MCI platform, the shopping-list game) were built for Assamese-speaking users or Assamese cultural content — we are the adaptation layer, not the inventors of the underlying task science.

Fixed-vocabulary voice interaction as an accessibility layer — using ASR only to recognize small, per-screen expected answers (rather than attempting general Assamese speech understanding, which isn't reliable yet) is our own practical design decision, built specifically around the real, published accuracy limitations of Assamese ASR.

Silent background cognitive-trend logging (via our companion module, Echo) feeding a caregiver dashboard — turning ordinary gameplay into a longitudinal engagement/progress signal — is our own integration, layered on top of validated game mechanics, not something inherited from any single cited source.

Part 6 — The Top 2 Games for MVP

Given realistic hackathon build time, here are the two games we recommend building first, and why:

1. "Ki Utsav?" (Game 1 — Episodic Memory / Reminiscence)

Why this one first: It is the simplest to build technically (image + question + 3 picture-choice answers), has the strongest, most direct research precedent (reminiscence-therapy apps are an entire proven commercial category), and it doubles as your best demo piece — cultural content is immediately visually compelling to judges and clearly shows the "built for Assam" story at a glance.

2. "Loga Loi Jua" (Game 6 — Daily Routine Recall / Sequencing)

Why this one second: It directly and explicitly answers the problem statement's named requirement ("daily routine recall"), has strong research grounding (the cooking-scenario studies), and demonstrates a different interaction pattern (drag/sequence, not just tap-to-choose) — showing judges you've built more than one type of mechanic, not just one game reskinned.

Together, these two cover two different cognitive domains, two different interaction styles, and both map directly and explicitly onto the problem statement's own wording — the strongest, most defensible MVP scope for the time available.

Document prepared as part of the SmritiSetu gaming module design. All research citations reference publicly available published studies, clinical trial registrations, or documented commercial serious-game precedents found during project research — full source list available on request.